



Factorization

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Factorization

$$12xy \checkmark = 3 \times 2 \times 2 \times x \times y \checkmark$$

$$24x^2y \checkmark = 2 \times 2 \times 2 \times 3 \times x \times x \times y$$

① Take the common out.

$$12xy + 24x^2y$$

$$3 \times 2 \times 2 \times x \times y (1 + 2x)$$

$$\Rightarrow 12xy (1 + 2x)$$

$$6x^2y^3 = \cancel{1} \times \underline{2} \times \underline{3} \times \underline{x} \times \underline{x} \times \underline{y} \times \underline{y} \times \underline{y}$$

$$\underline{18} x y^2 = \cancel{1} \times \underline{3} \times \underline{3} \times \underline{2} \times \underline{x} \times \underline{y} \times \underline{y}$$

$$4 \quad 2 \times 3 \times x \times y \times y (xy + 3)$$

$$= 6xy^2(xy + 3)$$

$$\begin{array}{r} 12xy + 12x^2y \\ \hline 12xy(1 + x) \end{array}$$

$$(ii) 4x^2y + 8x^2y^3 + 16xy$$

$$\boxed{2 \times 2 \times x \times x \times y} + \boxed{2 \times 2 \times 2 \times x \times x \times y \times y \times y} + \boxed{2 \times 2 \times 2 \times x \times y}$$

$$2 \times 2 \times x \times y [x + 2xy^2 + 4] = 4xy[x + 2xy^2 + 4]$$

$$12a^2b + 14ab$$

$$\downarrow$$

$$(2) \times 6 \times (a) \times a \times (b) + (2) \times 7 \times (a) \times (b)$$

$$2ab(6a + 7)$$

$$2((5x - 1)(x^2 + \dots))$$

Q.) $10x^2 - 14x^3 + 14x$

$$\cancel{2} \times \cancel{x} \times \cancel{5} - \cancel{2} \times \cancel{7} \times \cancel{x} \times \cancel{x} \times \cancel{x} + 3 \times 3 \times (x)(x)$$

$$22y - 33z = 11(2y - 3z)$$

$$6xy - 4x + 6 - 9x$$

$$\begin{aligned} & (6xy - 9x - 4x + 6) \\ & 3x(2y - 3) + 2(-2x + 3) \end{aligned}$$

$$2xy + 3x + 2y + 3$$

$$\begin{aligned} & 2xy + 2y + 3x + 3 \\ & 2y(x+1) + 3(x+1) \\ & \underline{\hspace{1cm}} \\ & \rightarrow (x+1)(2y+3) \end{aligned}$$

① Grouping Method

$$\begin{array}{l|l} 6xy + 6 - 9x - 4x & \\ \hline 6(xy+1) - x(9+4) & \end{array}$$

$$\begin{array}{l|l} 6xy - 4x + 6 - 9x & \\ \hline 6(xy+1) - x(4+9) & \\ \hline x^2 + xy + 8x + 8y & \\ & (6+x)(x+y) \end{array}$$

$$\begin{aligned}
 & (15xy) - 6x + (5y) - 2 \\
 & (5y-2)(3x+1) \quad \downarrow \quad \begin{array}{l} 1 \\ -2 \end{array}
 \end{aligned}$$

$5y = (5 \times y)$
 $15xy = 3 \times (5 \times x \times y)$
 $-6x = (2 \times 3 \times x)$
 $5y(3x+1) - 2(3x+1)$
 $(3x+1)(5y-2)$